

SMART INDIA HACKATHON 2026



DIODE GUARD

Problem Statement ID - SIH26145

Problem Statement Title - AI-Based Detection of Cyber Threats in Unidirectional IP Traffic

Theme – Blockchain & Cyber Security

PS Category – Software

Team ID –

Team Name – The S.H.I.E.L.D



Develop a passive threat detection pipeline by operating strictly behind one-directional hardware data diodes and analysing encrypted flow metadata through a fused Tri-Engine ML core, Diode Guard identifies both known volumetric attacks and novel zero-day botnet structures with sub-5ms latency and zero operational attack surface.

Proposed Solution

Passive Ingress & Air-Gapped Monitoring: Ingests network flow telemetry in a strictly unidirectional, read-only monitoring enclave using hardware data diodes. Ensuring zero risk of lateral movement or network tampering.

Payload-Agnostic Feature Extraction: Evaluates traffic patterns using flow-level metadata without decrypting payloads effective on TLS 1.3/QUIC traffic.

Tri-Engine Hybrid ML Detection Core:

- **Heuristic Engine** : Real-time deterministic matching for high-velocity signature threats (DDoS, port scanning).
- **Isolation Forest** : Dynamically isolates statistical anomalies and unexpected multi-dimensional traffic spikes from benign baselines.
- **Graph Neural Network** : Constructs dynamic local interaction subgraphs via PyTorch Geometric to expose hidden topological threats.

Live SOC Visualization: Dispatches scored alerts and telemetry to a real-time React dashboard over Web Sockets with sub-50ms latency.

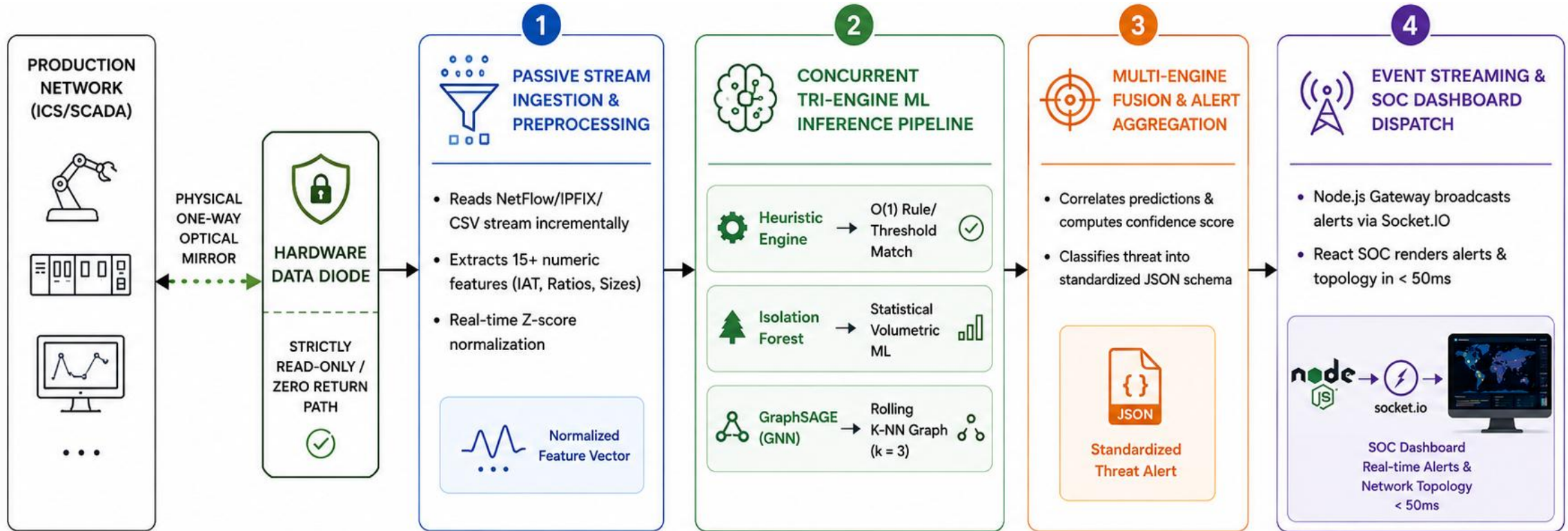
Innovation / Uniqueness

Fused Tri-Engine Detection: Combines the speed of deterministic rules, the novelty detection of unsupervised learning, and the structural intelligence of Graph Neural Networks (GNNs).

Graph Topological Modelling for Flows: Unlike conventional IDS that evaluate flows in isolation, Diode Guard models temporal topological relationships to unmask coordinated distributed botnets.

Hardware Data Diode Optimization: Built for critical infrastructure where active probing and traditional bidirectional firewalls introduce unacceptable vulnerability vectors.

Resource-Efficient & Line-Rate Scalable: Operates on lightweight numeric feature vectors rather than expensive packet reconstruction, enabling horizontal scaling at a fraction of enterprise appliance costs.



User Interface

React.js + Vite
Tailwind CSS
Lucid Icons

ML Frameworks

NumPy & Pandas
PyTorch & PyG
Scikit-Learn



Backend

Node.js & Socket.IO
REST Inference
Flask

Ingress Env.

Hardware Data Diode
NetFlow/IPFIX
Flow Probes

1. TECHNICAL

- Standard protocols: NetFlow, IPFIX, Zeek
- Lightweight ML: Scikit-Learn, PyTorch
- <5 ms inference per flow on commodity CPUs

3. ECONOMIC

- Uses numeric flow metadata instead of full packet inspection
- Eliminates costly proprietary DPI hardware
- Lower infrastructure & deployment cost



2. OPERATIONAL

- Plug-and-play deployment
- Works with optical taps / hardware data diodes
- Zero network configuration changes

4. SCALABILITY

- Docker/Kubernetes containerized architecture
- Supports horizontal scaling
- Can expand across server clusters

Challenges & Risk Area	Description & Threat Impact
1. Zero Payload Visibility	Pervasive end-to-end encryption blinds traditional DPI engines from inspecting packet content.
2. Strictly Read-Only	Hardware data diodes physically forbid two-way communication, preventing TCP resets, active probing, or challenge-response validation.
3. Real-Time Inference Bottlenecks	Deep Graph Neural Networks (GNNs) can introduce computational latency on high-throughput gigabit backbone lines.
4. False Positives & Concept Drift	Dynamic enterprise traffic shifts can trigger false alarms in unsupervised statistical models.

- **Zero Operational Disruption:** Ensures 24/7 continuous threat monitoring without risk of introducing latency, packet loss, or configuration instability into mission-critical ICS/SCADA systems.
 - **Immunity to Lateral Pivoting:** Even if external monitoring nodes are compromised, physically unidirectional data diodes prevent nation-state attackers from reaching production turbines, generators, or control systems
 - **Reduced Alert Fatigue:** Multi-engine fusion Rule + Isolation Forest + Graph SAGE prioritizes high-confidence structural threats over noisy single-packet false positives.
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- **Social & Public Safety Benefits:**
 - Civilian Infrastructure Protection:** Prevents catastrophic cyber-physical blackouts, communication outages, and municipal water contamination caused by cyber attacks.
 - **Economic & Financial Benefits:**
 - Loss Prevention:** Mitigates multi-million-dollar operational downtimes, regulatory penalties, and ransomware ransoms for national utilities.
 - **Strategic & National Security Benefits:**
 - Indigenous Cyber Defense:** Replaces reliance on foreign black-box proprietary security appliances with an open, auditable, high-performance AI detection pipeline.

➤ **Graph Neural Networks - GraphSAGE Architecture:**

Hamilton, W. L., Ying, R., & Leskovec, J. (2017). "Inductive Representation Learning on Large Graphs." Advances in Neural Information Processing Systems (NeurIPS).

Focus: Provides the mathematical foundation for dynamically embedding network flow topology without retraining on global graphs.

➤ **Unsupervised Anomaly Isolation:**

Liu, F. T., Ting, K. M., & Zhou, Z. H. (2008). "Isolation Forest." IEEE International Conference on Data Mining (ICDM).

Focus: Basis for fast, multi-dimensional statistical outlier isolation in high-throughput traffic.

➤ **Encrypted Malware & Behavioral Flow Analysis:**

Anderson, B., & McGrew, D. (Cisco Research, 2016). "Identifying Encrypted Malware Traffic with Contextual Flow Data." ACM Workshop on Artificial Intelligence and Security (AI-Sec).

Focus: Validates passive traffic fingerprinting using packet lengths and inter-arrival timing without TLS decryption.

➤ **Intrusion Dataset Benchmark & Feature Engineering:**

Sharafaldin, I., Lashkari, A. H., & Ghorbani, A. A. (2018). "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization." ICISSP.

Benchmark Datasets & Data Sources

CIC-IDS2017 / CSE-CIC-IDS2018 Dataset:

 [UNB Canadian Institute for Cybersecurity](#)

Real-world network flow telemetry covering DDoS, Botnet C2, Infiltration, Port Scanning, and Benign Baselines.

CTU-13 & Stratosphere Botnet Datasets:

 [Stratosphere IPS Project \(Czech Technical University\)](#)

Real-world botnet traffic captures for behavioural periodicity and C2 beaconing analysis.



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THANK YOU

